Netflix Customer, Subscription & Engagement Analytics

Using Dummy Customer Data

An End-to-End Data Analytics Project using Python, SQL & Power BI

1. Executive Summary

I have looked at customer data, subscription data, viewing data and engagement data to understand how customers interact with a Netflix-style streaming service and to find ways to improve customer retention, subscription strategy and content engagement.

I used an analytics workflow that starts with preparing and cleaning data in Python then uses SQL for business analysis and ends with Power BI visualizations. I focused on customer activity, subscription plans, revenue, account status, content preferences and customer engagement.

The final Power BI dashboard shows 983 customers, of which 697 are active. It also shows ₹72,118,471 in revenue an average of 42.69 watch hours, per customer and an average rating of 3.75 .

I see that the analysis gives a business view of the customer base and shows where to focus attention on retention, subscription management, content personalization and engagement.

2. Project Objectives

My main goal was to analyze Netflix dummy customer data and turn it into business insights about how customers behave what they subscribe to how they use content and how they stay engaged.

- I looked at the customer base to find which customers are active which have cancelled and which have paused.
- I studied subscription plans to see how customers are spread across plans and how money each plan brings in.
- I examined content preferences by genre and by content type.
- I measured how customers engage by counting watch hours, sessions and downloads.
- I compared how customers act in countries and, on different devices.

My overall goal was to use these insights to help decide how to keep customers set subscription plans recommend content and boost customer engagement.

3. Data & Methodology

3.1 Data Used

The project uses a customer viewing and subscription dataset containing **1,013 records and 20 columns**.

The main variables used in the analysis include:

Data Area	Variables
Customer	Customer ID, Customer Name, Age
Content	Title, Content Type, Genre
Location	Country
Subscription	Subscription Plan, Monthly Fee
Account	Account Status
Device	Device,Payment Method
Activity	Join Date, Last Watch Dat
Viewing	Watch Hours, Movies Watched, Episodes Watched
Engagement	Monthly Sessions, Download Count
Rating	Average Rating

3	CUST1063	Customer 631	38	Bridgerton	Movie	Documentary	Japan	Basic	Mobile	PayPal	Active
4	CUST1068	Customer 683	40	Money Heist	TV Show	Sci-Fi	France	Basic	Tablet	Debit Card	Paused
5	CUST1051	Customer 515	44	You	TV Show	Action	Spain	Premium	Gaming Cc	Credit Carc	Active
6	CUST1036	Customer 366	57	The Witcher	TV Show	Comedy	Canada	Premium	Smart TV	Credit Carc	Active
7	CUST1065	Customer 656	42	You	Movie	NULL	Germany	Basic	Mobile	Credit Carc	Paused
8	CUST1065	Customer 657	22	Squid Game	TV Show	Documentary	Canada	Premium	Smart TV	Credit Carc	Active
9	CUST1053	Customer 530	61	Elite	Movie	Drama	Germany	Standard	Tablet	Debit Card	Paused
10	CUST1032	Customer 322	24	Stranger Things	Movie	Horror	Canada	Standard	Smart TV	Credit Carc	Active
11	CUST1007	Customer 71	18	Elite	TV Show	Horror	United Ki	Standard	Mobile	PayPal	Paused
12	CUST1089	Customer 891	44	Bridgerton	Movie	Documentary	Canada	Basic	Mobile	PayPal	Cancelled
13	CUST1037	Customer 372	200	Black Mirror	TV Show	Documentary	Japan	Standard	Gaming Cc	Credit Carc	Active
14	CUST1076	Customer 768	60	Narcos	TV Show	Documentary	South Kore	Basic	Smart TV	UPI	Active
15	CUST1045	Customer 457	37	Black Mirror	TV Show	Drama	Canada	Basic	Tablet	Credit Carc	Active

Fig1. Raw Netflix Dummy Data

3.2 Data Preparation

Python was used to clean and prepare the raw dataset before analysis.

The main steps included:

- Checked the dataset structure, columns and data types.
- Identified **5 duplicate records** and removed them.
- Converted values such as `N/A`, `NULL` and blanks into missing values.
- Removed unnecessary spaces from text fields.
- Standardized fields such as **Country, Genre, Subscription Plan and Account Status**.
- Converted viewing, subscription and engagement fields into numeric formats.
- Identified invalid values in **Age, Watch Hours and Monthly Fee**.
- Removed invalid records before further analysis.
- Checked the cleaned dataset for remaining data-quality issues.

```
memory usage: 158.4+ KB

[10]: df['Join_Date'] = pd.to_datetime(df['Join_Date'], errors = 'coerce')
      df['Last_Watch_Date'] = pd.to_datetime(df['Last_Watch_Date'], errors = 'coerce')
```

```
# Removing duplicates , inplace makes sure it get rmeoved from og data
df.drop_duplicates(inplace=True)

df.duplicated().sum()

np.int64(0)

# Dirty Values - N/A, NULL , Blank , Spaces
df.replace(
    ['N/A','NULL','',' '],np.nan,inplace = True
)

df.isnull().sum()

# Extra Spaces Remvoing
df = df.apply(lambda x : x.str.strip() if x.dtype == 'object' else x)

#Standardize Text
#  Data can in upper /lower/mix in nay format so have to standardize it propely.

df = df.apply(lambda x: x.astype(str).str.strip().str.title() if x.dtype == 'object' else x)
```

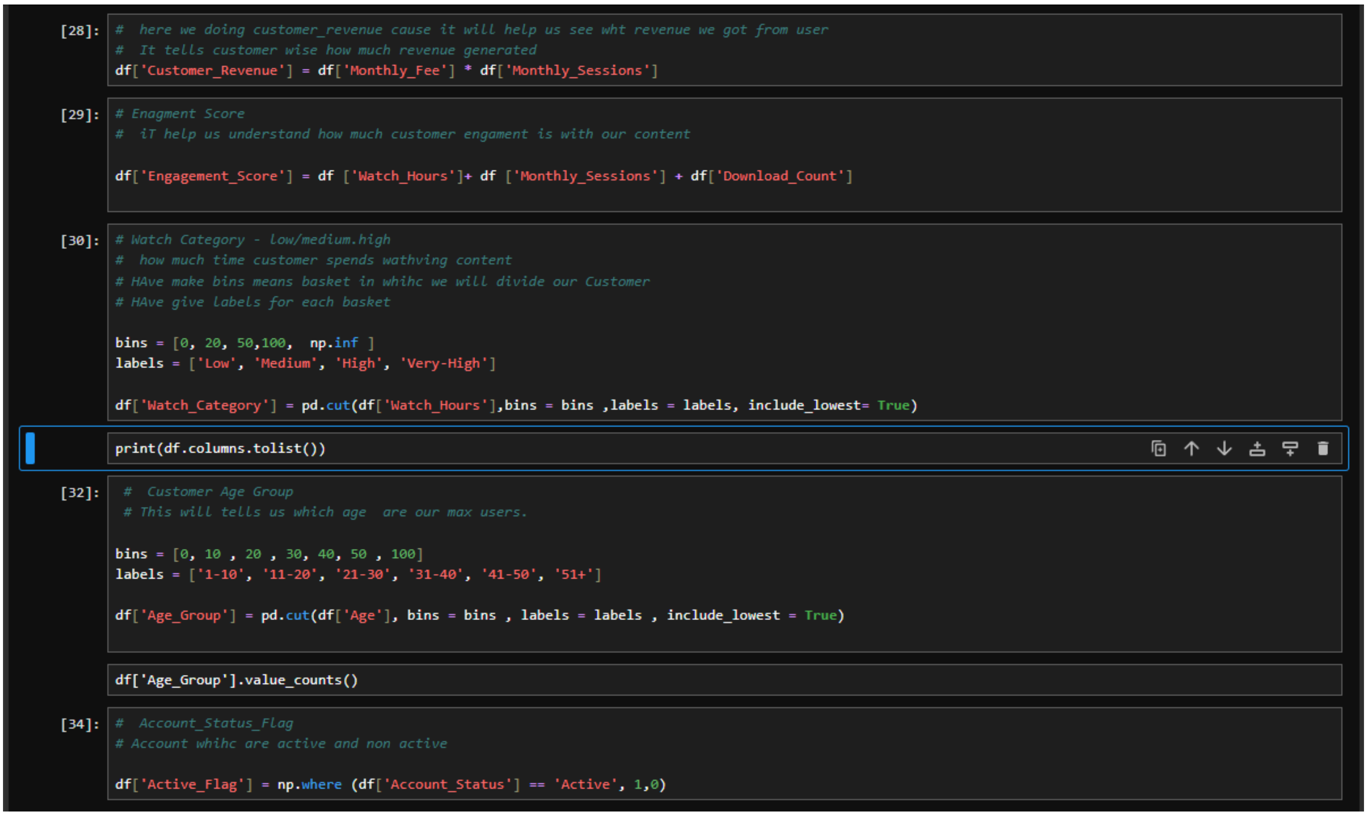
```
[21]: #  make varible of all numeric columns

      numeric_columns = ['Age', 'Watch_Hours', 'Monthly_Fee', 'Movies_Watched',
                         'Episodes_Watched','Avg_Rating', 'Monthly_Sessions',
                         'Download_Count']

[22]: # This will makse sure all numeric columns are numeric and erroe will marked as nan
      for column in numeric_columns :
        df[column] = pd.to_numeric(df[column],errors = 'coerce')

      df.describe(include = 'number')

[24]: # here above we can see that column age have negavtive in min and 200 somehting in max which is wrong so do this set a limit
      df = df[(df['Age']>= 1)& (df['Age']<=100)]

      df.describe(include = 'number')

[26]: #  Negative  Values
      df = df[df['Watch_Hours']>= 0]
      df = df[df['Monthly_Fee']>=0]

[27]: df.isnull().sum()
```

Figure 2. Data Transformation Using Python

3.3 Feature Engineering

Additional fields were created to support business analysis:

- **Customer Revenue** — Monthly Fee × Monthly Sessions
- **Engagement Score** — Watch Hours + Monthly Sessions + Download Count
- **Watch Category** — Low, Medium, High and Very-High
- **Age Group** — Divided in Bucket decade-based age bands.
- **Active Flag** — Active Account as 1 and all others as 0.



```
[28]: #  here we doing customer_revenue cause it will help us see wht revenue we got from user
      #  It tells customer wise how much revenue generated
      df['Customer_Revenue'] = df['Monthly_Fee'] * df['Monthly_Sessions']

[29]: # Engagment Score
      #  iT help us understand how much customer engament is with our content

      df['Engagement_Score'] = df ['Watch_Hours']+ df ['Monthly_Sessions'] + df['Download_Count']

[30]: # Watch Category - low/medium.high
      #  how much time customer spends wathving content
      # HAve make bins means basket in whihc we will divide our Customer
      # HAve give labels for each basket

      bins = [0, 20, 50,100,  np.inf ]
      labels = ['Low', 'Medium', 'High', 'Very-High']

      df['Watch_Category'] = pd.cut(df['Watch_Hours'],bins = bins ,labels = labels, include_lowest= True)

      print(df.columns.tolist())

[32]:  #  Customer Age Group
       # This will tells us which age  are our max users.

      bins = [0, 10 , 20 , 30, 40, 50 , 100]
      labels = ['1-10', '11-20', '21-30', '31-40', '41-50', '51+']

      df['Age_Group'] = pd.cut(df['Age'], bins = bins , labels = labels , include_lowest = True)

      df['Age_Group'].value_counts()

[34]: #  Account_Status_Flag
      # Account whihc are active and non active

      df['Active_Flag'] = np.where (df['Account_Status'] == 'Active', 1,0)
```

Figure 3. Feature Engineering Using Python

3.4 Analytical Tools

- Python was used for data cleaning, preparation, feature engineering and exploratory analysis.
- SQL was then used to answer business questions around customers, subscription plans, revenue, genres, countries, devices, watch hours, content type, ratings and account status.
- Power BI was used to convert the results into an interactive dashboard containing KPI cards, customer analysis, content analysis and engagement visuals.
- Workflow:
- Raw Data → Python → SQL → Power BI → Insights → Recommendations

4. Key Findings & Visualizations

Finding 1 — Customer Account Status

We can see three customer-status segments:

- Active — 697 customers
- Cancelled — 197 customers
- Paused — 89 customers.

Business meaning :

1. Customer Retention

Customer Retention is about keeping the 697 active customers interested in our content. Customer Retention means sending them updates and encouraging them to keep using the platform. Customer Retention also means listening to their feedback and adding features that they love.

2. Reactivation & Churn

Reactivation & Churn gives us a chance to bring back the 89 customers. Reactivation & Churn also asks us to look at the 197 cancelled customers so we can find out why they left. Reactivation & Churn is, about learning from their stories and preventing cancellations.

3. Engagement Monitoring

Engagement Monitoring lets us watch how customers watch and interact with our services. Engagement Monitoring helps us spot customers whose activity is falling. Engagement Monitoring can alert us before those customers become inactive.

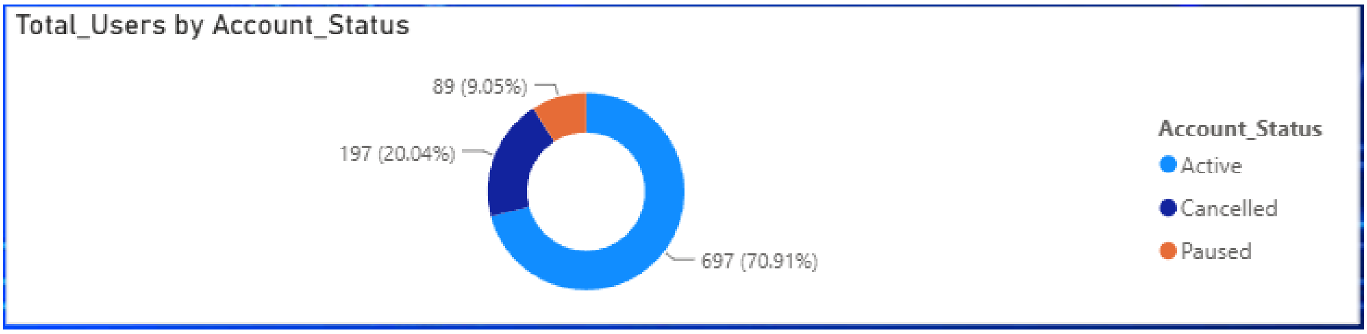


Figure 4. Customer Account Status Distribution

Finding 2 — Subscription Plans

The dashboard shows customers across the Premium, Standard and Basic subscription plans. Premium has highest the customers, then Standard, then Basic.

Business Meaning :

1. Premium Leads the Customer Base:

Premium plan has 341 customers, which makes Premium plan the popular subscription plan. Standard plan follows with 326 customers while Basic plan has 316 customers.

2. Customer Distribution Is Relatively Balanced :

I notice that the difference between Premium plan and Basic plan is 25 customers showing that customers are fairly evenly distributed across all three subscription tiers instead of being heavily concentrated in one plan.

3. Premium Shows Stronger Customer Preference :

I see that Premium plan, with 341 customers holds the share of the customer base. This shows that customers prefer the higher-tier subscription more, than Standard plan and Basic plan.

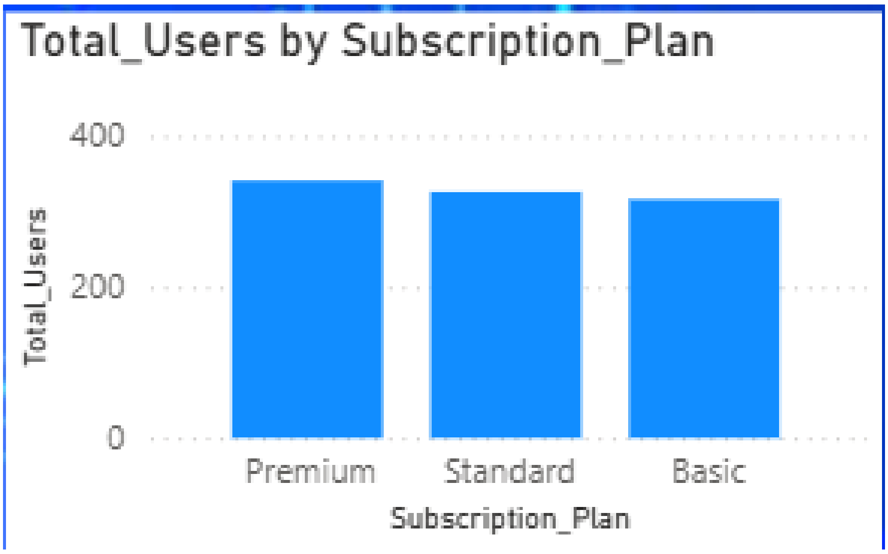


Figure 5. Customer Distribution by Subscription Plan

Finding 3 — Average Watch Hours by Genre

The analysis shows that **Horror has the highest average watch time at 45 hours**, followed closely by **Sci-Fi at 44.91 hours**. Action and Drama also show strong viewing levels at **43.36** and **42.94 hours** respectively.The lowest average watch hours are for **Thriller (41.58 hours)** and **Crime (41.68 hours)**.

Business Meaning:

1. Horror & Sci-Fi Have Stronger Engagement

Customers spend the most time watching **Horror (45 hours)** and **Sci-Fi (44.91 hours)**, indicating stronger viewing engagement with these genres.

2. Viewing Time Is Fairly Consistent

Average watch hours range from **41.58 to 45 hours**, a difference of only **3.42 hours**. This shows that customer viewing time is relatively consistent across genres.

3. Genre Performance Can Guide Content Focus

The higher watch time for Horror and Sci-Fi indicates stronger engagement in these categories, while the lower viewing time for Thriller and Crime can be analyzed further to understand differences in customer interest and content consumption.

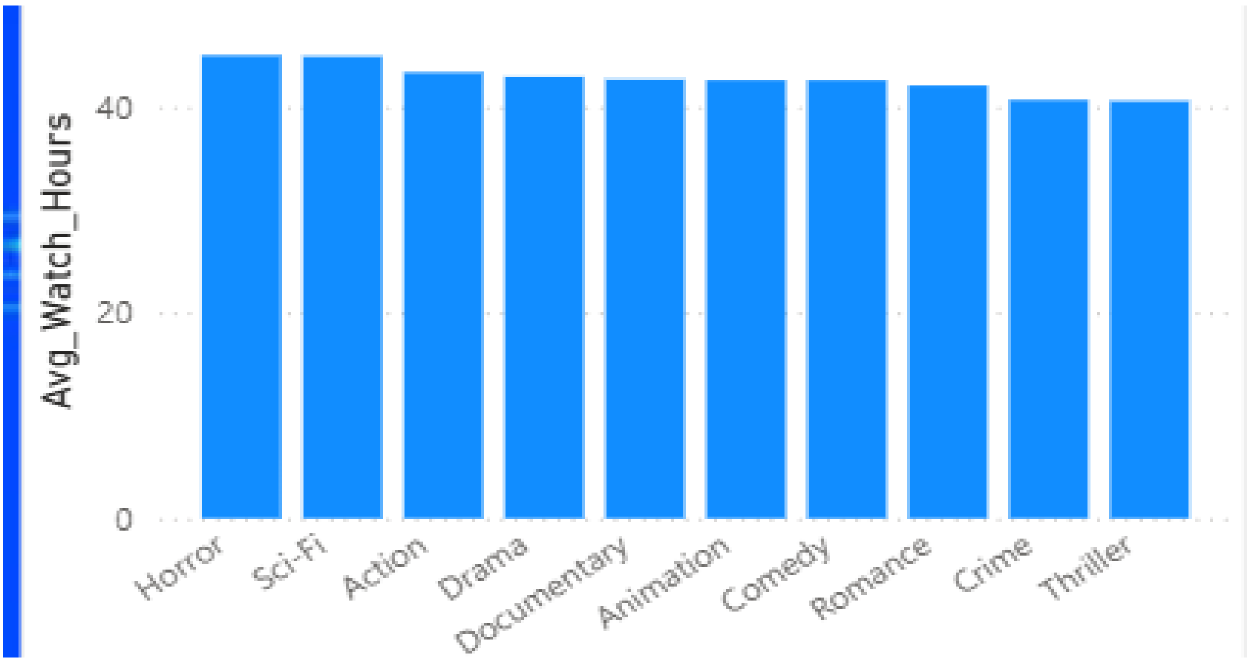


Figure 6. Average Watch Hours by Genre

Finding 4 —Average Watch Hours Decline Over Time

The analysis shows that average watch hours decreased consistently from 2022 to 2025.

2022 → 44.47 hours

2023 → 42.45 hours

2024 → 41.97 hours

2025 → 41.70 hours

Business Meaning :

1. Customer Viewing Has Declined

Average watch hours went down from 44.47 hours in 2022 to 41.70 hours in 2025. This shows that customers are spending time watching content.

2. The Rate of Decline Is Slowing

The drop was 2.02 hours from 2022 to 2023. Only 0.48 hours from 2023 to 2024 and 0.27 hours from 2024 to 2025. This shows that the decline is getting smaller over time.

3. Engagement Should Be Monitored

The ongoing drop in watching hours means that customer engagement needs to be watched. Looking at this trend along with genre, subscription plan and content preferences can help see what is causing the changes, in watching habits

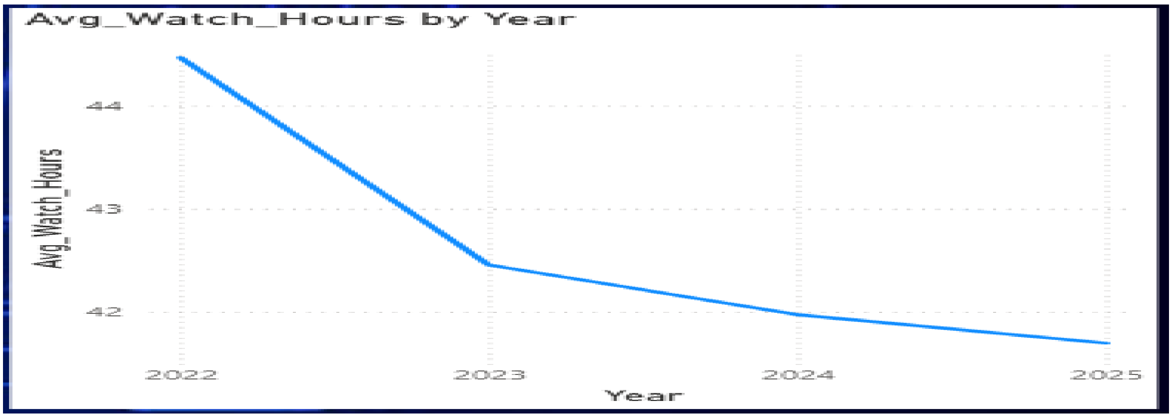


Figure 7. Average Watch Hours by Year

Finding 5 — TV Shows Have Slightly Higher Usage Than Movies

The analysis shows that **507 users watch TV Shows**, while **476 users watch Movies**. TV Shows therefore have **31 more users** than Movies, showing a slight preference for series-based content.

Business Meaning :

1. TV Shows Have Higher Usage

TV Shows have **507 users (51.6%)**, compared with **476 users (48.4%)** for Movies, showing a slightly stronger preference for TV content.

2. Customer Preference Is Balanced

The difference of only **31 users** indicates that customers consume both content types, rather than having a strong preference for only Movies or TV Shows.

3. Content Variety Matters

Since usage is relatively balanced, offering a good mix of **TV Shows and Movies** remains important for meeting different customer viewing preferences.

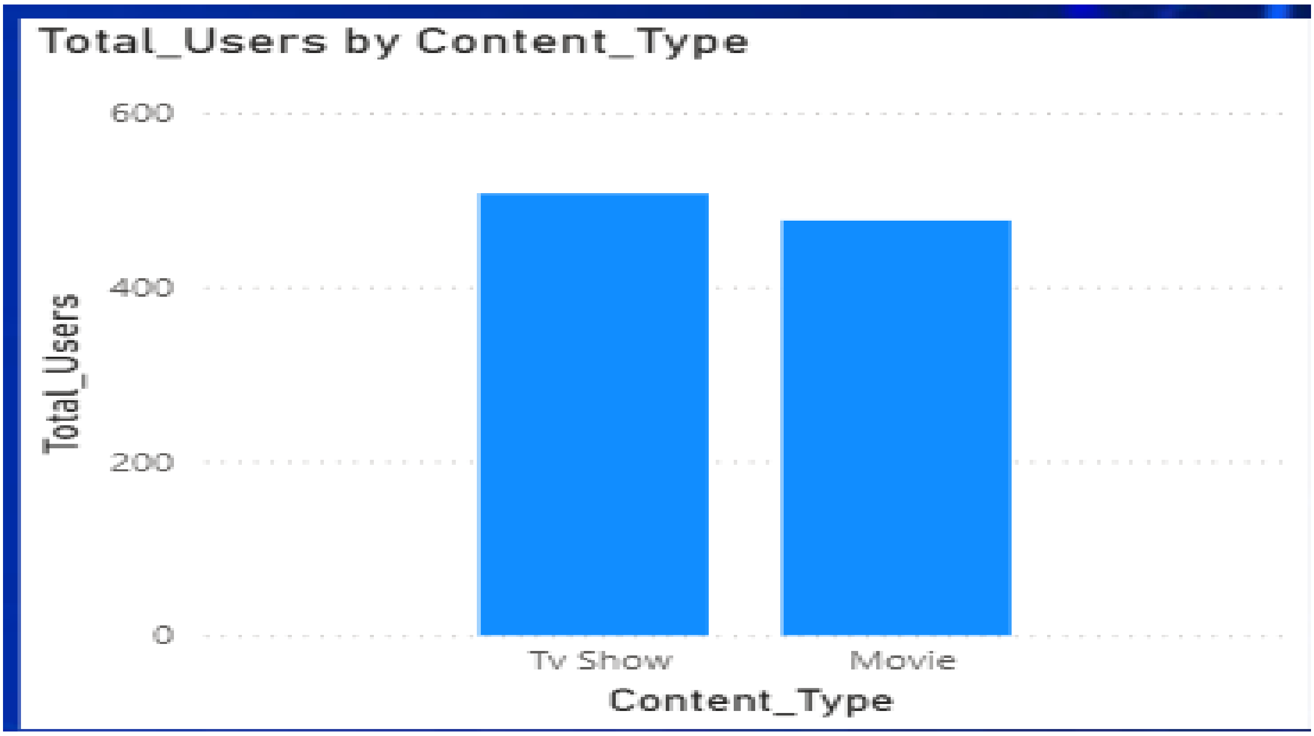


Figure 8. Total Users by Content Type

5. PowerBI Dashboard

The final Power BI dashboard combines the KPIs and visualizations from the analysis into one business view.

This dashboard gives a view of:

1. Overall Performance: Total Users, Active Users, Total Revenue, Average Watch Hours and Average Rating
2. Customer Analysis: Users by Subscription Plan, Country and Account Status
3. Content Analysis: Average Watch Hours by Genre and Users by Content Type
4. Engagement Trends: Average Watch Hours by Year
5. Interactive Filters: Device and Age Group
6. It allows users to explore data dynamically and gain insights, through filters and visual representations.

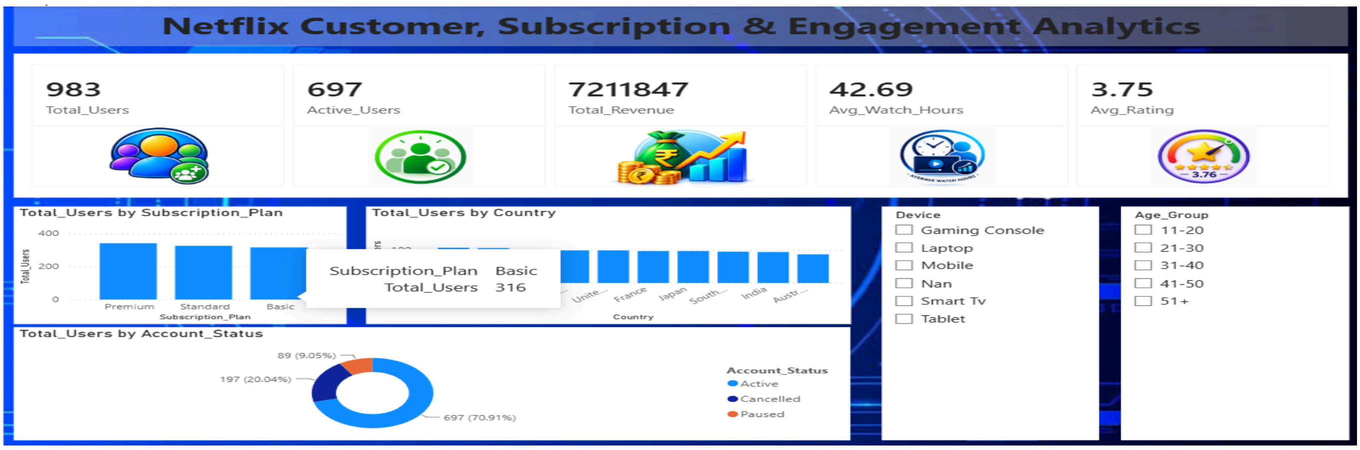


Figure 9. Final PowerBI Dashboard - page1

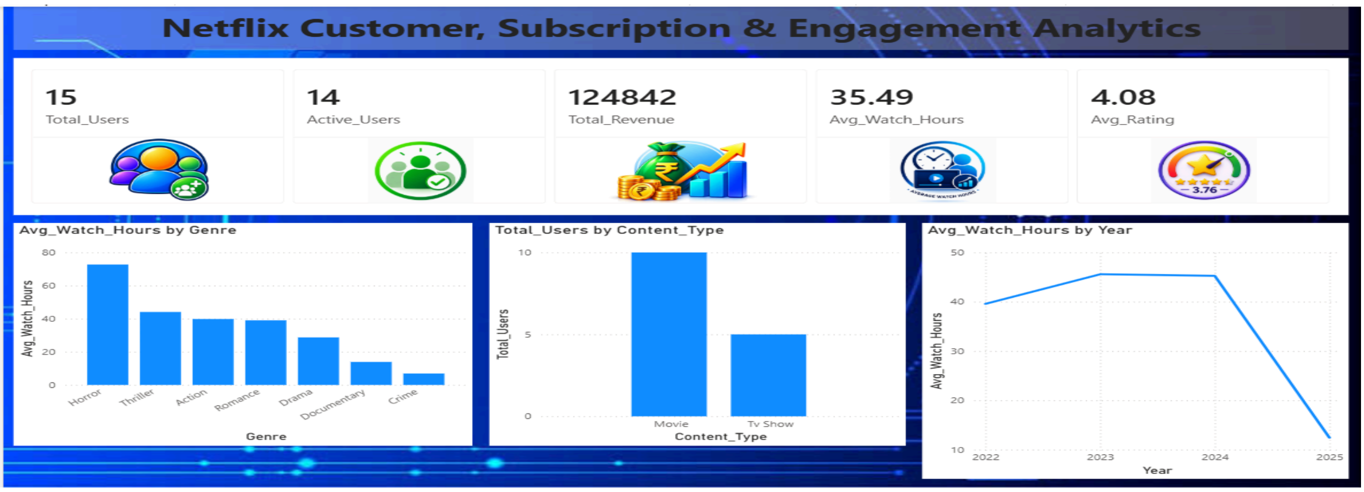


Figure 10. Final PowerBI Dashboard - page2

6. Actionable Recommendations

1. Customer Retention & Reactivation

Customer Retention & Reactivation must focus on keeping the 697 customers involved by providing content that matches their interests and encouraging them to use the platform regularly. Customer Retention & Reactivation must also try to bring the 89 customers who have stopped using the service by offering them special deals and content that fits their tastes.

The 197 customers who have stopped their subscriptions need to be looked at based on how they used their subscription and what they watched. This will help business figure out why they left. Knowing why customers leave can help business stop people from leaving in the future.

2. Premium Upgrade Strategy

Premium Upgrade Strategy needs to focus on the 341 customers who have the Premium subscription. These customers are the group compared to the 326 customers with Standard and the 316 customers with Basic.

The business can find customers who watch a lot and are very active among the Standard and Basic groups by looking at how hours they spend watching and what they watch. Then business can offer them deals to upgrade to Premium. This can help more people choose the Premium option and bring in money from the people already using the service.

3. High-Engagement Genre Strategy

High-Engagement Genre Strategy must focus on the genres that people watch the most. Horror is the top with 45 hours. Sci-Fi is next with 44.91 hours.

The business can make sure that content from these genres is easy to find by putting it in a spot on the homepage and suggesting it to customers. At the time genres that don’t do as well like Thriller with 41.58 hours and Crime with 41.68 hours can be studied to see why people are not watching as much. This can help business understand what customers like better.

4. Customer Engagement & Watch-Time Strategy

Customer Engagement & Watch-Time Strategy must deal with the drop in average watch hours from 44.47 hours in 2022 to 41.70 hours, in 2025.

The business can check how hours people are watching across different genres, different subscription levels and different types of content to see where the drop is happening. Then business can give recommendations and better ways to find content to customers who are watching less. This can help business keep people interested and stop the drop in watching from getting worse.

5. Balanced Movies & TV Shows Strategy

Balanced Content Strategy must keep both Movies and TV Shows available because people watch them the same amount. TV Shows have 507 users, which's 51.6 percent while Movies have 476 users, which is 48.4 percent. The difference is 31 users.

The business can keep offering a mix of Movies and TV Shows and use what each person watches to decide which to suggest to them. This can help meet viewing choices and make it easier for people to find the content they like.

7. Conclusion

I have seen that customer behavior is influenced by account status, subscription plan, genre, viewing time and content type.

The customer analysis identified 697 customers, 89 paused customers and 197 cancelled customers. This creates opportunities for customer retention, reactivation and churn reduction. The subscription analysis shows that Premium has the customer base with 341 customers, followed by Standard with 326 customers and Basic with 316 customers. This gives an opportunity to increase Premium adoption.

The content analysis shows engagement with Horror at 45 hours and Sci-Fi at 44.91 hours. The average watch hours declined from 44.47 hours in 2022 to 41.70 hours in 2025. The content-type analysis also shows a preference, between TV Shows, used by 507 users and Movies used by 476 users.

Overall the project converts customer data into business insights and actionable recommendations. These cover customer retention, reactivation, subscription growth, content engagement, viewing activity and content strategy.